

Stat 400: Applied Probability and Statistics I

Spring 2025—Sections: 0311, 0312, 0313, 0321, 0322, 0323, 0331, 0332, 0333.

Instructor: Sana Jahedi

Office: Atlantic Building room 4313

Lectures: TuTh 9:30pm - 10:45pm, ARM 0135

Email: sjahedi@umd.edu (Always Include Stat 400 in Email Subject)

Office Hours:

- Thursdays 11:00 -12:20 in person.
- Thursday 3:30-4:30 in person
- Tuesdays 5:00-6:00 Zoom, use the link zoom office hours. Tuesday office hours will begin from Feb 11.

Textbooks: I will use the following text books in this course.

- **Introduction to Probability and Statistics, Dr. D. Gunatilleka and Dr. J. Fernandes.**
- Probability and Statistics: For Engineering and the Science, 9th Edition, Jay L. Devore. (Chapters 2-6)
- Mathematical Statistics and Data Analysis, Third Edition, John A. Rice (Chapters 1-5)
- Introduction to probability Second Edition, J Blitzstein and J Hwang,
- Introduction to Probability Models, Sheldon M. Ross. (Chapters 1-3)

The main book that I will follow is the first one in the above list, book by Gunatilleka and Fernandes, which will be in canvas in Files/Readings-BookChapters. The book by J Blitzstein and J Hwang is available for free, to access you can click on the link provided above. You do not have to buy any of the other books. From time to time, I may recommend you to read a few pages of one of the above books. In that case, I will upload a scan of those pages to LMS Canvas; you will be able to find it under Files/Readings-BookChapters.

Teacher Assistants Information:

Sec	Discussion	TA	Email	Office*	Office hours
0311, 0321, 0331	MTH 0307	Mariam Askari	fmaskari@umd.edu	MTH 2119	F 12-1 pm
0312, 0322, 0332	PHY 2213	Jiajing Guan	jguan4@umd.edu	MTH 4400	F 1-2 pm
0313, 0323, 0333	MTH 0403	Soyoung Park	soyoung@umd.edu	PHY 2113	F 1-2 pm

Important Links:

- Academic deadlines
- Course-Related Policies and Resources for Undergraduate Students
- Student Rights and Responsibilities
- Accessibility and Disability Services. If you have a diagnosed disability please make sure to request accommodations by registering with Accessibility and Disability Services, and let me know that I can help you in the best way possible.
- The Counseling Center
- The Office of Civil Rights and Sexual Misconduct

Piazza: I have set up UMD-Piazza for this class. Feel free to use the platform to communicate with the rest of your classmates. You can ask questions and share your ideas about your homeworks, course materials, and practice problems, but complete solution to a homework problem is not allowed. If you are shy, do not worry you can ask a question anonymously. Please be respectful and kind towards each other, we all are learning together.

Sign up for your Piazza forum [here](#)

Resources available for you:

- You have access to past final exams [here](#). Check out Stat400.
- Free tutoring is available in Math Building room 0203, see [here](#). It has free tutoring for both R programming and Stat400.

- You can also get free help from tutors of the math success program. Check the Math Success Program
- There is also counseling services available for math anxiety, preparation, study skills, etc. You can schedule an academic coaching session here.

Course Scope:

Prerequisites: Informally speaking Calculus 2. Formally speaking (MATH131 or MATH141).

The major topics that we will cover in this course are:

- Experiment, Sample Space, Event
- Axioms of probability, counting and calculating probabilities
- Conditional probability and Bay's Theorem
- Independence
- Random variables; both discrete and continuous
- Expected value, Variance and Standard Deviation - Examples of discrete random variables
- Examples of continuous random variables
- Joint probability distribution
- Properties of expected values, Covariance and correlation
- Statistics and Sample mean distribution
- Point Estimation - Other Topics; for example Confidence intervals (if time permits).
- Simulations

Deep, meaningful learning requires sustained effort on your part. Certain learning techniques are known to be much more effective than others (self-testing is much more efficient and effective than re-reading and highlighting). The catch is that it requires more discipline and effort. You will probably spend from 2-3 hours working outside the class for every hour in class. That is you should be spending 6-9 hours each week on homework and understanding outside of class time. If you fall behind it will be very hard for you to catch up so you should plan your term wisely.

Attendance Policy:

- **ATTENDANCE IN YOUR TUTORIAL SESSIONS IS MANDATORY.** There will be biweekly Quizzes which will take place in your discussion section. There will be no make up Quiz, except for university approved absence. In the weeks you do not have quizzes

there will be discussion worksheets which includes examples. You should attend your discussion sections and work on your worksheets, your TAs will go over the questions.

- Attendance in lecture is not mandatory, however the lectures will not be recorded. I may use some simple explanations during lectures that help you learn the concept better. In addition, there will be some bonus questions (extra credit) that will be given during lectures, so if you do not attend the lectures you will miss the bonuses. In general, active participation in the class is highly appreciated. You can always teach new things to me and your friends! Please do your best to attend all the lectures!

Homework: There will be biweekly assignment that will be posted on Tuesday and will be due to Sun 11:59 p.m. I will post your homework on the Gradescope. You should show all your works and submit your answer to the grade scope. The purpose is to make sure you know how to write an answer with complete steps. I would suggest always keep a hard copy of your homework (including your solution for the multiple choice one), in that case it will be easier for self study later and also if you have a question regarding your homework, I would ask you to come to the office hours with a hard copy of your work. **The lowest homework will be dropped.**

Quiz/Labs: Quizzes are 30% of whole your grade. The exact content of a quiz will be announced two sessions in advance. **there will be no make up Quiz, the only exception is an Approved Excused Absence.** For university policy on excused absence click here. **The lowest Quiz will be dropped.**

Tests: There will be two midterms. Including the Final Exam, all the Tests have the same weight towards your final grade. **Tests, including Midterms and the Final Exam will count towards your final grade, there will be no make up test, the only exception is an Approved Excused Absence.** For university policy on excused absence click here.

Grading	
Homework*	25
Quiz*	30
Midterm 1	15
Midterm 2	15
Final Exam	15
Total Score	100

* Please do your best to submit all your assignments and be prepared for all your quizzes!
The lowest homework grade and the lowest quiz grade will be dropped at the end of the semester.

Note : The cutoffs for this course are given below.

	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	D-
Cut-off	97.0	91	89	88	80	79.0	78.0	70.0	69.0	68.0	60.0	59.0

Tentative Semester Schedule: NOTE: The following schedule is subject to change!
for the updated Schedule look at the google doc, [click here](#) Exact dates and content of Tests will be announced at least two weeks in advance.

Week	Tue	Th	Fri	Sun
Jan 28th–31th	Ch 2.1. Sample Space and Events	Ch 2.2. Axioms and Properties of Probability (a)	Discussion 1	
Feb 4th–7th	Ch 2.2. Axioms and Properties of Probability (b)	Ch 2.3. Counting Techniques	Discussion 2	HW 1 DDL
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Week	Tue	Th	Fri	Sun
Feb 11th–14th	Ch 2.4. Conditional Probability and Bays Thm. (a)	Ch 2.4. Conditional Probability and Bays Thm. (b)	Quiz 1: Sec 2.1, 2.2, 2.3	
Feb 18th–21th	Ch 2.5. Independence	Ch 3.1. Random variables Ch 3.2 Discrete random variables and Probability Distributions	Discussion 3	HW 2 DDL
Feb 25th–28th	Ch 4.1. Continuous Random Variables Ch 4.2 (a) Cumulative Distribution Functions Probability Distribution	Ch 3.3 Expected Value, Variance and Standard Deviation	Quiz 2: TBA	
Mar 4th–7th	Ch 3.4 The Binomial Probability Distribution Ch 4.2 (b) Expected Value	Ch 3.5 Geometric and Negative Binomial	Discussion 4	HW 3 DDL
Mar 11th–14th	Review	Midterm 1	Review on integration methods: basic double integration	
Mar 18th–21th	March Break	March Break	March Break	
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Week	Tue	Th	Fri	Sun
Mar 25th– 28th	Ch 3.6 Poisson distribution	Ch 4.3 Normal Random Variables	Discussion 5	HW 4 DDL
Apr 1st–4th	Ch 4.4 Exponential distribution Preparation for Chapter 5, reviewing double integrals	Ch 5.1 Jointly Distributed Random Variables (a)	Quiz 3: TBA	
Apr 8th–11th	Ch 5.1 Jointly Distributed Random Variables (b)	Ch 5.2 Expected Values, Covariance	Quiz 4: TBA	HW 5 DDL
Apr 15th–18th	Ch 5.5. Linearity of expectation and correlation	Function of random variables	Review	HW 6 DDL
Apr 22nd–25th	Review	Midterm 2	No discussion section	
Apr 29th– May 2nd	Ch 5.3. Statistics and their distributions	Ch 5.4. The Central Limit	Discussion 6	HW 7 DDL
May 6th–9th	Ch 6.1 Point Estimation (a)	Ch 6.2 Point Estimation (b)	Quiz 5: TBA	
May 13th– 16th	Review (last session)		Final exam 10:30 a.m. - 12:30 p.m.	